GRADE 12

MATHEMATICS

CHAPTER 3

Analytical Solid Geometry

Coordinates, lines, planes and spheres in three-dimensional space

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Chapter Outline

Analytical Solid Geometry

3.1 Coordinates of a Point in Space

Equation of the Axes

Axis	Equation	Coordinate Form
x-axis	$y = 0,\ z = 0$	$(x, 0, 0)$
y-axis	$x = 0,\ z = 0$	$(0, y, 0)$
z-axis	$x = 0,\ y = 0$	$(0, 0, z)$

Equation of the Coordinate Planes

Coordinate Plane	Equation	Coordinate Form
xy-plane	$z = 0$	$(x, y, 0)$
yz-plane	$x = 0$	$(0, y, z)$
zx-plane	$y = 0$	$(x, 0, z)$

Planes Parallel to Coordinate Planes

Plane	Equation	Coordinate Form
Parallel to xy-plane	$z = c$	(x, y, c)
Parallel to yz-plane	$x = a$	(a, y, z)
Parallel to zx-plane	$y = b$	(x, b, z)

Lines Perpendicular to Coordinate Planes

Line	Equation	Coordinate Form
Perpendicular to xy-plane	$x = a,\ y = b$	(a, b, z)
Perpendicular to yz-plane	$y = b,\ z = c$	(x, b, c)
Perpendicular to zx-plane	$x = a,\ z = c$	(a, y, c)

Distance Between Two Points

If $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are two points in space, then the distance between P and Q is

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example 1

Find the equation of the line through the point $(-3, 5, 7)$ and perpendicular to
(a) xy-plane **(b)** yz-plane **(c)** zx-plane.
Find the point of intersection of each line and plane.

Solution

(a) The equation of the line through the point $(-3, 5, 7)$ and perpendicular to the xy-plane is

$$x = -3, \ y = 5 \qquad \text{or} \qquad (-3, 5, z).$$

The point of intersection of the line and the xy-plane is $(-3, 5, 0)$.

(b) The equation of the line through the point $(-3, 5, 7)$ and perpendicular to the yz-plane is

$$y = 5, \ z = 7 \qquad \text{or} \qquad (x, 5, 7).$$

The point of intersection of the line and the yz-plane is $(0, 5, 7)$.

(c) The equation of the line through the point $(-3, 5, 7)$ and perpendicular to the zx-plane is

$$x = -3, \ z = 7 \qquad \text{or} \qquad (-3, y, 7).$$

The point of intersection of the line and the zx-plane is $(-3, 0, 7)$.

Exercise 3.1

1. Find the equation of the line through the point $(2, 3, -4)$ and perpendicular to
(a) xy-plane **(b)** yz-plane **(c)** zx-plane.
Find the point of intersection of the line and plane.

Solution

(a) The equation of the line through the point $(2, 3, -4)$ and perpendicular to the xy-plane is

$$x = 2, \ y = 3 \qquad \text{or} \qquad (2, 3, z).$$

The point of intersection of the line and the xy-plane is $(2, 3, 0)$.

(b) The equation of the line through the point $(2, 3, -4)$ and perpendicular to the yz-plane is

$$y = 3, \ z = -4 \qquad \text{or} \qquad (x, 3, -4).$$

The point of intersection of the line and the yz-plane is $(0, 3, -4)$.

(c) The equation of the line through the point $(2, 3, -4)$ and perpendicular to the zx-plane is

$$x = 2, \ z = -4 \qquad \text{or} \qquad (2, y, -4).$$

The point of intersection of the line and the zx-plane is $(2, 0, -4)$.

2. Find the equation of the plane containing the point $(1,-2,3)$ and parallel to
(a) xy-plane **(b)** yz-plane **(c)** zx-plane.

Solution

(a) The equation of the plane containing the point $(1,-2,3)$ and parallel to the xy-plane is

$$z = 3.$$

(b) The equation of the plane containing the point $(1,-2,3)$ and parallel to the yz-plane is

$$x = 1.$$

(c) The equation of the plane containing the point $(1,-2,3)$ and parallel to the zx-plane is

$$y = -2.$$

3. Find the distance between the points $(2,-3,5)$ and $(7,5,-2)$.

Solution

Let P be the point $(2,-3,5)$ and Q be the point $(7,5,-2)$.

$$\begin{aligned} PQ &= \sqrt{(7-2)^2 + \big(5-(-3)\big)^2 + (-2-5)^2} \\ &= \sqrt{5^2 + 8^2 + (-7)^2} \\ &= \sqrt{25+64+49} \\ &= \sqrt{138}. \end{aligned}$$

Therefore, the distance between the two points is $\sqrt{138}$.

4. Show that the points $(-1,2,5)$, $(1,1,6)$ and $(0,5,6)$ form a right triangle.

Solution

Let A be the point $(-1,2,5)$, B be the point $(1,1,6)$ and C be the point $(0,5,6)$.

$$\begin{aligned} AB^2 &= (1-(-1))^2 + (1-2)^2 + (6-5)^2 = 4+1+1 = 6, \\ BC^2 &= (0-1)^2 + (5-1)^2 + (6-6)^2 = 1+16+0 = 17, \\ AC^2 &= (-1-0)^2 + (2-5)^2 + (5-6)^2 = 1+9+1 = 11. \end{aligned}$$

Since $AB^2 + AC^2 = 6 + 11 = 17 = BC^2$, the triangle is right-angled at A.
Hence, the points A, B and C form a right triangle.

5. Show that the points $(1,-1,2)$, $(3,-2,3)$ and $(5,-3,4)$ are collinear.

Solution

Let A be the point $(1,-1,2)$, B be the point $(3,-2,3)$ and C be the point $(5,-3,4)$.

$$\begin{aligned} AB &= \sqrt{(3-1)^2 + \big(-2-(-1)\big)^2 + (3-2)^2} = \sqrt{4+1+1} = \sqrt{6}, \\ BC &= \sqrt{(5-3)^2 + \big(-3-(-2)\big)^2 + (4-3)^2} = \sqrt{4+1+1} = \sqrt{6}, \\ AC &= \sqrt{(5-1)^2 + \big(-3-(-1)\big)^2 + (4-2)^2} = \sqrt{16+4+4} = 2\sqrt{6}. \end{aligned}$$

Since $AB + BC = \sqrt{6} + \sqrt{6} = 2\sqrt{6} = AC$, the point B lies on the line segment joining A and C.
Therefore, the points A, B and C are collinear.

3.2 Lines

Directed values of a line segment

Directed values of a line segment PQ are

$$\langle PQ \rangle = \langle l, m, n \rangle = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Length of the line segment PQ is

$$|PQ| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} = \sqrt{l^2 + m^2 + n^2}$$

Equation of the line

Coordinate form

$$(x, y, z) = (x_1 + kl, y_1 + km, z_1 + kn)$$

Symmetric form

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} \quad (l \neq 0,\ m \neq 0,\ n \neq 0)$$

Example 2

Given $P(1,2,3)$ and $Q(3,6,5)$, find the coordinates of the point $R(x,y,z)$ on the line PQ with respect to the point P for the following values of k.

(a) $k=\dfrac{1}{2}$ (b) $k=2$ (c) $k=-2$

Solution

$$\langle PQ\rangle = \langle 3-1, 6-2, 5-3\rangle = \langle 2,4,2\rangle.$$

Coordinates of the points on the line PQ are

$$(x,y,z) = (1+2k, 2+4k, 3+2k).$$

(a) When $k=\dfrac{1}{2}$,

$$R(x,y,z) = \left(1+2\cdot\tfrac{1}{2},\ 2+4\cdot\tfrac{1}{2},\ 3+2\cdot\tfrac{1}{2}\right) = (2,4,4).$$

(b) When $k=2$,

$$R(x,y,z) = (1+2\cdot 2,\ 2+4\cdot 2,\ 3+2\cdot 2) = (5,10,7).$$

(c) When $k=-2$,

$$R(x,y,z) = (1+2\cdot(-2),\ 2+4\cdot(-2),\ 3+2\cdot(-2)) = (-3,-6,-1).$$

Exercise 3.2

1. Given $P(3,1,5)$ and $Q(-3,7,-2)$, find the coordinates of the point $R(x,y,z)$ on the line PQ with respect to the point P for the following values of k.

(a) $k=\dfrac{1}{2}$ (b) $k=3$ (c) $k=-2$

Solution

$$\langle PQ\rangle = \langle -3-3, 7-1, -2-5\rangle = \langle -6,6,-7\rangle.$$

Coordinates of the points on the line PQ are

$$(x,y,z) = (3-6k, 1+6k, 5-7k).$$

(a) When $k=\dfrac{1}{2}$,

$$R(x,y,z) = \left(3-6\cdot\tfrac{1}{2},\ 1+6\cdot\tfrac{1}{2},\ 5-7\cdot\tfrac{1}{2}\right) = \left(0,4,\tfrac{3}{2}\right).$$

(b) When $k=3$,

$$R(x,y,z) = (3-6\cdot 3,\ 1+6\cdot 3,\ 5-7\cdot 3) = (-15,19,-16).$$

(c) When $k=-2$,

$$R(x,y,z) = (3-6\cdot(-2),\ 1+6\cdot(-2),\ 5-7\cdot(-2)) = (15,-11,19).$$

Example 3

Given $P(-1,2,3)$ and $Q(3,5,-2)$, determine whether or not the following points are on the line PQ. If the point is on the line PQ, find the corresponding parameter with respect to the point P.

(a) $\left(1,\dfrac{7}{2},\dfrac{1}{2}\right)$ (b) $(7,8,-7)$ (c) $(-5,-1,8)$ (d) $(7,8,-2)$

Solution

$$\langle PQ\rangle = \langle 3-(-1), 5-2, -2-3\rangle = \langle 4,3,-5\rangle.$$

Coordinates of the points on the line PQ are

$$(x,y,z) = (-1+4k, 2+3k, 3-5k) \text{ for some real number } k.$$

(a) If $(x,y,z) = \left(1, \frac{7}{2}, \frac{1}{2}\right)$, then

$$1 = -1 + 4k \qquad \frac{7}{2} = 2 + 3k \qquad \frac{1}{2} = 3 - 5k$$
$$k = \frac{1}{2} \qquad k = \frac{1}{2} \qquad k = \frac{1}{2}$$

Therefore, the point $\left(1, \frac{7}{2}, \frac{1}{2}\right)$ is on the line PQ and the corresponding parameter is $k = \frac{1}{2}$.

(b) If $(x,y,z) = (7,8,-7)$, then

$$7 = -1 + 4k \qquad 8 = 2 + 3k \qquad -7 = 3 - 5k$$
$$k = 2 \qquad k = 2 \qquad k = 2$$

Therefore, the point $(7,8,-7)$ is on the line PQ and the corresponding parameter is $k = 2$.

(c) If $(x,y,z) = (-5,-1,8)$, then

$$-5 = -1 + 4k \qquad -1 = 2 + 3k \qquad 8 = 3 - 5k$$
$$k = -1 \qquad k = -1 \qquad k = -1$$

Therefore, the point $(-5,-1,8)$ is on the line PQ and the corresponding parameter is $k = -1$.

(d) If $(x,y,z) = (7,8,-2)$, then

$$7 = -1 + 4k \qquad 8 = 2 + 3k \qquad -2 = 3 - 5k$$
$$k = 2 \qquad k = 2 \qquad k = 1$$

Since the values of k are not equal, the point $(7,8,-2)$ is not on the line PQ.

Example 4

Given $P(2,1,3)$ and $Q(6,-5,3)$, determine whether or not the following points are on the line PQ. If the point is on the line PQ, find the corresponding parameter with respect to the point P.
(a) $(4,-2,3)$ (b) $(-2,7,3)$ (c) $(10,-11,3)$ (d) $(1,1,3)$

Solution

$$\langle PQ \rangle = \langle 6-2, -5-1, 3-3 \rangle = \langle 4,-6,0 \rangle.$$

Coordinates of the points on the line PQ are

$$(x,y,z) = (2+4k, 1-6k, 3) \text{ for some real number } k.$$

(a) If $(x,y,z) = (4,-2,3)$, then

$$4 = 2 + 4k \qquad -2 = 1 - 6k$$
$$k = \frac{1}{2} \qquad k = \frac{1}{2}$$

Therefore, the point $(4,-2,3)$ is on the line PQ and the corresponding parameter is $k = \frac{1}{2}$.

(b) If $(x,y,z) = (-2,7,3)$, then

$$-2 = 2 + 4k \qquad 7 = 1 - 6k$$
$$k = -1 \qquad k = -1$$

Therefore, the point $(-2,7,3)$ is on the line PQ and the corresponding parameter is $k = -1$.

(c) If $(x, y, z) = (10, -11, 3)$, then

$$10 = 2 + 4k \qquad -11 = 1 - 6k$$
$$k = 2 \qquad k = 2$$

Therefore, the point $(10, -11, 3)$ is on the line PQ and the corresponding parameter is $k = 2$.

(d) If $(x, y, z) = (1, 1, 3)$, then

$$1 = 2 + 4k \qquad 1 = 1 - 6k$$
$$k = -\frac{1}{4} \qquad k = 0$$

Since the values of k are not equal, the point $(1, 1, 3)$ is not on the line PQ.

Exercise 3.2

2. Given $P(-2, 1, 3)$ and $Q(4, 4, -3)$, determine whether or not the following points are on the line PQ. If the point is on the line PQ, find the corresponding parameter with respect to the point P.
(a) $(6, 3, -6)$ **(b)** $(6, 5, -5)$ **(c)** $(-4, 0, 5)$ **(d)** $(7, 8, -2)$

Solution

$$\langle PQ \rangle = \langle 4 - (-2), 4 - 1, -3 - 3 \rangle = \langle 6, 3, -6 \rangle.$$

Coordinates of the points on the line PQ are

$$(x, y, z) = (-2 + 6k, 1 + 3k, 3 - 6k) \text{ for some real number } k.$$

(a) If $(x, y, z) = (6, 3, -6)$, then

$$6 = -2 + 6k \qquad 3 = 1 + 3k \qquad -6 = 3 - 6k$$
$$k = \frac{4}{3} \qquad k = \frac{2}{3} \qquad k = \frac{3}{2}$$

Since the values of k are not equal, the point $(6, 3, -6)$ is not on the line PQ.

(b) If $(x, y, z) = (6, 5, -5)$, then

$$6 = -2 + 6k \qquad 5 = 1 + 3k \qquad -5 = 3 - 6k$$
$$k = \frac{4}{3} \qquad k = \frac{4}{3} \qquad k = \frac{4}{3}$$

Therefore, the point $(6, 5, -5)$ is on the line PQ and the corresponding parameter is $k = \frac{4}{3}$.

(c) If $(x, y, z) = (-4, 0, 5)$, then

$$-4 = -2 + 6k \qquad 0 = 1 + 3k \qquad 5 = 3 - 6k$$
$$k = -\frac{1}{3} \qquad k = -\frac{1}{3} \qquad k = -\frac{1}{3}$$

Therefore, the point $(-4, 0, 5)$ is on the line PQ and the corresponding parameter is $k = -\frac{1}{3}$.

(d) If $(x, y, z) = (7, 8, -2)$, then

$$7 = -2 + 6k \qquad 8 = 1 + 3k \qquad -2 = 3 - 6k$$

$$k = \frac{3}{2} \qquad k = \frac{7}{3} \qquad k = \frac{5}{6}$$

Since the values of k are not equal, the point $(7, 8, -2)$ is not on the line PQ.

3. Given $P(3, 2, -1)$ and $Q(4, 2, 5)$, determine whether or not the following points are on the line PQ. If the point is on the line PQ, find the corresponding parameter with respect to the point P.

(a) $(5, 2, 11)$ **(b)** $(2, 2, -7)$ **(c)** $\left(\frac{7}{2}, 2, 2\right)$ **(d)** $(6, 2, 10)$

Solution

$$\langle PQ \rangle = \langle 4 - 3, 2 - 2, 5 - (-1) \rangle = \langle 1, 0, 6 \rangle.$$

Coordinates of the points on the line PQ are

$$(x, y, z) = (3 + k, 2, -1 + 6k) \text{ for some real number } k.$$

(a) If $(x, y, z) = (5, 2, 11)$, then

$$5 = 3 + k \qquad 11 = -1 + 6k$$

$$k = 2 \qquad k = 2$$

Therefore, the point $(5, 2, 11)$ is on the line PQ and the corresponding parameter is $k = 2$.

(b) If $(x, y, z) = (2, 2, -7)$, then

$$2 = 3 + k \qquad -7 = -1 + 6k$$

$$k = -1 \qquad k = -1$$

Therefore, the point $(2, 2, -7)$ is on the line PQ and the corresponding parameter is $k = -1$.

(c) If $(x, y, z) = \left(\frac{7}{2}, 2, 2\right)$, then

$$\frac{7}{2} = 3 + k \qquad 2 = -1 + 6k$$

$$k = \frac{1}{2} \qquad k = \frac{1}{2}$$

Therefore, the point $\left(\frac{7}{2}, 2, 2\right)$ is on the line PQ and the corresponding parameter is $k = \frac{1}{2}$.

(d) If $(x, y, z) = (6, 2, 10)$, then

$$6 = 3 + k \qquad 10 = -1 + 6k$$

$$k = 3 \qquad k = \frac{11}{6}$$

Since the values of k are not equal, the point $(6, 2, 10)$ is not on the line PQ.

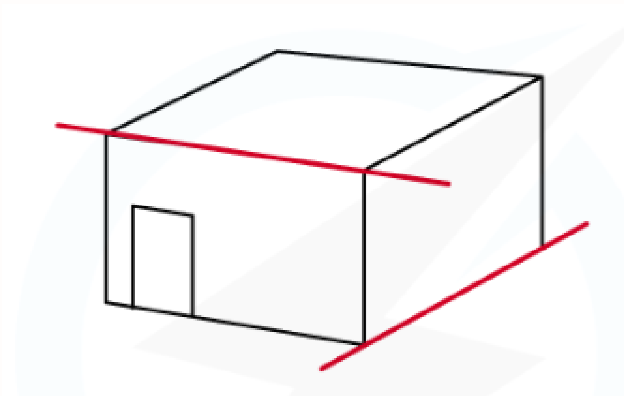
3.3 Parallel, Skew and Perpendicular Lines

Parallel Lines

$$\langle PQ \rangle = \langle OA \rangle \Leftrightarrow PQ \parallel OA \text{ and } PQ = OA$$
$$\langle PQ \rangle = k\langle OA \rangle \Leftrightarrow PQ \parallel OA$$

Skew Lines

In space, there are pairs of lines that are neither parallel nor intersect. These pairs of lines are called skew lines.



Finding the Angle Between Two Lines

$$\cos \angle PAQ = \frac{l_1 l_2 + m_1 m_2 + n_1 n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2} \cdot \sqrt{l_2^2 + m_2^2 + n_2^2}}$$
$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0 \Rightarrow \cos \angle PAQ = 0 \Rightarrow \angle PAQ = 90^\circ$$

Perpendicular Lines

Two lines are perpendicular if and only if they intersect and $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$ for any directed values $\langle l_1, m_1, n_1 \rangle$ and $\langle l_2, m_2, n_2 \rangle$ of the lines.

Example 5

Given $P(2,1,3)$, $Q(6,-5,4)$, $R(2,3,4)$ and $S(-1,5,1)$, determine whether the lines PQ and RS are parallel or skew or intersect.

Solution

$$\langle PQ\rangle = \langle 6-2, -5-1, 4-3\rangle = \langle 4,-6,1\rangle$$
$$\langle RS\rangle = \langle -1-2, 5-3, 1-4\rangle = \langle -3,2,-3\rangle$$

Since $\langle PQ\rangle$ is not a scalar multiple of $\langle RS\rangle$, the lines PQ and RS are not parallel.
Coordinates of the points on the lines are

$$PQ: \ (x,y,z) = (2+4s, 1-6s, 3+s)$$
$$RS: \ (x,y,z) = (2-3t, 3+2t, 4-3t)$$

If the two lines intersect, then they have a common point. So,

$$2+4s = 2-3t \qquad 1-6s = 3+2t \qquad 3+s = 4-3t$$
$$4s = -3t \qquad -6s-2 = 2t \qquad s+3t = 1$$

From $4s=-3t$, we get $t = -\dfrac{4s}{3}$. Substituting into $s+3t=1$,

$$s + 3\left(-\frac{4s}{3}\right) = 1$$
$$-3s = 1$$
$$s = -\frac{1}{3}, \qquad t = \frac{4}{9}$$

Substituting these values into $-6s-2=2t$, we get $0 \neq \dfrac{8}{9}$.
This is impossible. Therefore, the lines PQ and RS do not intersect.
Hence, the lines PQ and RS are **skew lines**.

Example 6

Given $P(0,0,1)$, $Q(3,6,4)$, $R(0,3,1)$ and $S(3,0,4)$, show that the lines PQ and RS are perpendicular.

Solution

$$\langle PQ\rangle = \langle 3-0, 6-0, 4-1\rangle = \langle 3,6,3\rangle$$
$$\langle RS\rangle = \langle 3-0, 0-3, 4-1\rangle = \langle 3,-3,3\rangle$$

Coordinates of the points on the lines are

$$PQ: (x,y,z) = (3s, 6s, 1+3s)$$
$$RS: (x,y,z) = (3t, 3-3t, 1+3t)$$

If the two lines intersect, then their coordinates must be equal. So,

$$3s = 3t \qquad 6s = 3-3t \qquad 1+3s = 1+3t$$
$$s = t \qquad 2s+t = 1 \qquad s = t$$

Substituting $t = s$ into $2s + t = 1$,

$$2s + s = 1$$
$$3s = 1$$
$$s = \frac{1}{3} \qquad t = \frac{1}{3}$$

Therefore, the two lines intersect at the point $(x, y, z) = (1, 2, 2)$.

$$l_1l_2 + m_1m_2 + n_1n_2 = (3)(3) + (6)(-3) + (3)(3) = 9 - 18 + 9 = 0$$

Since the lines PQ and RS intersect and $l_1l_2 + m_1m_2 + n_1n_2 = 0$, the lines PQ and RS are **perpendicular**.

Example 7

Find the equation of the line passing through the point $(-4, 7, -3)$ and perpendicular to the line $(x, y, z) = (3 + 2k, -1 + 3k, 1 - k)$. Find also the point of intersection of the two lines.

Solution

Directed values of the given line are $\langle 2, 3, -1\rangle$.

Directed values of the required line are

$$\langle -4 - (3 + 2k),\ 7 - (-1 + 3k),\ -3 - (1 - k)\rangle = \langle -7 - 2k,\ 8 - 3k,\ -4 + k\rangle$$

for some real number k.

If two lines are perpendicular, then

$$2(-7 - 2k) + 3(8 - 3k) + (-1)(-4 + k) = 0$$
$$-14 - 4k + 24 - 9k + 4 - k = 0$$
$$14k = 14$$
$$k = 1$$

So directed values of the required line are $\langle -9, 5, -3\rangle$ and the equation of the line is

$$(x, y, z) = (-4 - 9t, 7 + 5t, -3 - 3t)$$

The point of intersection is

$$(x, y, z) = (5, 2, 0).$$

Exercise 3.3

1. Find $\cos\angle PAQ$ for the following.

(a) $P(1, 2, -1),\ A(-2, 1, 5),\ Q(2, -1, 0)$

(b) $P(0, 2, -3),\ A(2, -1, 5),\ Q(-2, 3, -1)$

Solutions

(a) $\langle AP\rangle = \langle 3, 1, -6\rangle, \quad \langle AQ\rangle = \langle 4, -2, -5\rangle$

$$\cos\angle PAQ = \frac{(3)(4) + (1)(-2) + (-6)(-5)}{\sqrt{3^2 + 1^2 + (-6)^2}\cdot\sqrt{4^2 + (-2)^2 + (-5)^2}} = \frac{40}{\sqrt{46}\cdot\sqrt{45}} = \frac{40}{3\sqrt{230}} \approx 0.8792$$

(b) $\langle AP\rangle = \langle -2, 3, -8\rangle, \quad \langle AQ\rangle = \langle -4, 4, -6\rangle$

$$\cos\angle PAQ = \frac{(-2)(-4) + (3)(4) + (-8)(-6)}{\sqrt{(-2)^2 + 3^2 + (-8)^2}\cdot\sqrt{(-4)^2 + 4^2 + (-6)^2}} = \frac{68}{\sqrt{77}\cdot\sqrt{68}} = \frac{34}{\sqrt{1309}} \approx 0.9397$$

2. Determine whether the lines PQ and RS are parallel or skew or intersect. If PQ and RS intersect, are they perpendicular?
(a) $P(1,2,3),\ Q(4,5,6),\ R(-2,3,5),\ S(4,9,11)$
(b) $P(3,-1,-3),\ Q(2,-3,1),\ R(3,-2,5),\ S(-1,-2,1)$
(c) $P(4,-2,5),\ Q(-2,6,1),\ R(-1,1,4),\ S(3,3,2)$
(d) $P(-3,-1,6),\ Q(-1,3,0),\ R(0,6,7),\ S(-4,-4,-1)$

Solutions

(a) $\langle PQ\rangle = \langle 3,3,3\rangle$ and $\langle RS\rangle = \langle 6,6,6\rangle$
$\langle PQ\rangle = \frac{1}{2}\langle RS\rangle$
Therefore, the lines PQ and RS are **parallel**.

(b) $\langle PQ\rangle = \langle -1,-2,4\rangle$ and $\langle RS\rangle = \langle -4,0,-4\rangle$. Since these are not scalar multiples, the lines are not parallel.
Coordinates of the points on the lines are

$$PQ : (x,y,z) = (3-s, -1-2s, -3+4s)$$
$$RS : (x,y,z) = (3-4t, -2, 5-4t)$$

If the two lines intersect, then their coordinates must be equal. So,

$$-1-2s = -2 \qquad s = \frac{1}{2}$$
$$3-s = 3-4t \qquad t = \frac{1}{8}$$
$$-3+4s = 5-4t \qquad -1 \neq \frac{9}{2}$$

This is impossible. Therefore, the lines PQ and RS do not intersect.
Hence, they are **skew lines**.

(c) $\langle PQ\rangle = \langle -6,8,-4\rangle$ and $\langle RS\rangle = \langle 4,2,-2\rangle$. Since these are not scalar multiples, the lines are not parallel.
Coordinates of the points on the lines are

$$PQ:\ (x,y,z) = (4-6s, -2+8s, 5-4s)$$
$$RS:\ (x,y,z) = (-1+4t, 1+2t, 4-2t)$$

If the two lines intersect, then their coordinates must be equal. So,

$$4-6s = -1+4t \qquad 5-6s = 4t$$
$$-2+8s = 1+2t \qquad -3+8s = 2t$$
$$5-4s = 4-2t \qquad t = 2s-\frac{1}{2}$$

Substituting $t = 2s - \frac{1}{2}$ into $-3+8s = 2t$,

$$\begin{aligned} -3+8s &= 2\left(2s-\frac{1}{2}\right) \\ -3+8s &= 4s-1 \\ 4s &= 2 \\ s &= \frac{1}{2}, \qquad t = \frac{1}{2} \end{aligned}$$

Therefore, the two lines intersect at the point $(x,y,z) = (1,2,3)$.

$$\begin{aligned} l_1l_2 + m_1m_2 + n_1n_2 &= (-6)(4) + (8)(2) + (-4)(-2) \\ &= -24 + 16 + 8 \\ &= 0 \end{aligned}$$

Since the lines intersect and $l_1l_2 + m_1m_2 + n_1n_2 = 0$, the lines PQ and RS are **perpendicular**.

(d) $\langle PQ \rangle = \langle 2, 4, -6 \rangle$ and $\langle RS \rangle = \langle -4, -10, -8 \rangle$.

Since these are not scalar multiples, the lines are not parallel.

Coordinates of the points on the lines are

$$\begin{aligned} PQ : (x, y, z) &= (-3 + 2s, -1 + 4s, 6 - 6s) \\ RS : (x, y, z) &= (-4t, 6 - 10t, 7 - 8t) \end{aligned}$$

If the two lines intersect, then their coordinates must be equal. So,

$$\begin{aligned} -3 + 2s &= -4t \\ t &= \frac{3 - 2s}{4} \end{aligned} \qquad \begin{aligned} -1 + 4s &= 6 - 10t \\ 4s + 10t &= 7 \end{aligned} \qquad 6 - 6s = 7 - 8t$$

Substituting $t = \dfrac{3 - 2s}{4}$ into $4s + 10t = 7$,

$$\begin{aligned} 4s + 10\left(\frac{3 - 2s}{4}\right) &= 7 \\ 8s + 15 - 10s &= 14 \\ -2s &= -1 \\ s &= \frac{1}{2}, \qquad t = \frac{1}{2} \end{aligned}$$

Therefore, the two lines intersect at the point $(x, y, z) = (-2, 1, 3)$.

$$\begin{aligned} l_1l_2 + m_1m_2 + n_1n_2 &= (2)(-4) + (4)(-10) + (-6)(-8) \\ &= -8 - 40 + 48 \\ &= 0 \end{aligned}$$

Since the lines intersect and $l_1l_2 + m_1m_2 + n_1n_2 = 0$, the lines PQ and RS are **perpendicular**.

3. Find the equation of the line passing through the point $(8, -1, -10)$ and perpendicular to the line $(x, y, z) = (1 + 2k, 2 - k, 3 - 7k)$. Find also the point of intersection of the two lines.

Solution

Directed values of the given line are $\langle 2, -1, -7 \rangle$.

Directed values of the required line are

$$\langle 8 - (1 + 2k),\ -1 - (2 - k),\ -10 - (3 - 7k) \rangle = \langle 7 - 2k,\ -3 + k,\ -13 + 7k \rangle$$

for some real number k.

If two lines are perpendicular, then

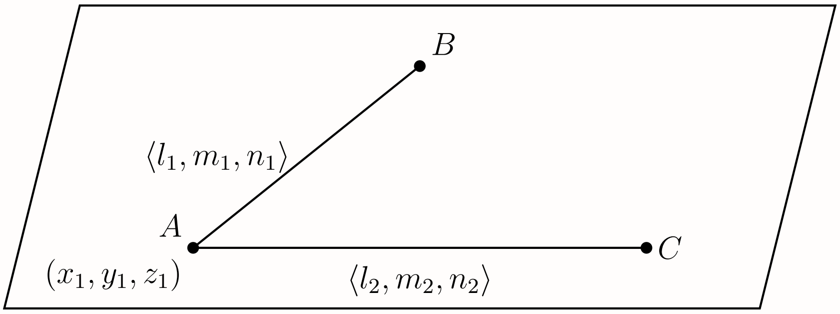
$$\begin{aligned} 2(7 - 2k) + (-1)(-3 + k) + (-7)(-13 + 7k) &= 0 \\ 14 - 4k + 3 - k + 91 - 49k &= 0 \\ 108 - 54k &= 0 \\ k &= 2 \end{aligned}$$

So directed values of the required line are $\langle 3, -1, 1 \rangle$ and the equation of the line is

$$(x, y, z) = (8 + 3t, -1 - t, -10 + t)$$

The point of intersection is $(x, y, z) = (5, 0, -11)$.

3.4 Planes



Equation of a Plane

$\langle AB\rangle = \langle l_1, m_1, n_1\rangle$ and $\langle AC\rangle = \langle l_2, m_2, n_2\rangle$
$a = m_1n_2 - m_2n_1$
$b = n_1l_2 - n_2l_1$
$c = l_1m_2 - l_2m_1$
$d = ax_1 + by_1 + cz_1$
Cartesian form of the plane equation

$$ax + by + cz = d$$

Example 8

Find the equation of the plane containing $A(1, 0, 1)$, $B(3, 6, 4)$ and $C(-2, 3, 1)$.

Solution

$\langle AB\rangle = \langle 2, 6, 3\rangle, \qquad \langle AC\rangle = \langle -3, 3, 0\rangle$

Hence,

$$\begin{aligned}
a &= m_1n_2 - m_2n_1 = 6(0) - 3(3) = -9\\
b &= n_1l_2 - n_2l_1 = 3(-3) - 0(2) = -9\\
c &= l_1m_2 - l_2m_1 = 2(3) - (-3)(6) = 24
\end{aligned}$$

Using the point $A(1, 0, 1)$,

$$\begin{aligned}
-9x - 9y + 24z &= d\\
d &= (-9)(1) + (-9)(0) + (24)(1) = 15
\end{aligned}$$

Therefore, the equation of the plane is

$$\begin{aligned}
-9x - 9y + 24z &= 15\\
3x + 3y - 8z &= -5
\end{aligned}$$

Exercise 3.4

1. Find the equation of the plane containing
(a) $A(2,-5,4)$ and $B(-5,2,4)$ and $C(-2,3,-1)$
(b) $A(4,2,-3)$ and $B(1,-2,4)$ and $C(-1,0,3)$

Solutions

(a) $\langle AB\rangle = \langle -7,7,0\rangle$ and $\langle AC\rangle = \langle -4,8,-5\rangle$.
Hence,

$$a = 7(-5) - 8(0) = -35$$
$$b = 0(-4) - (-5)(-7) = -35$$
$$c = (-7)(8) - (-4)(7) = -28$$

Using the point $A(2,-5,4)$,

$$-35x - 35y - 28z = d$$
$$d = (-35)(2) + (-35)(-5) + (-28)(4) = -7$$

Therefore, the equation of the plane is

$$-35x - 35y - 28z = -7$$
$$5x + 5y + 4z = 1$$

(b) $\langle AB\rangle = \langle -3,-4,7\rangle$ and $\langle AC\rangle = \langle -5,-2,6\rangle$.
Hence,

$$a = (-4)(6) - (-2)(7) = -10$$
$$b = 7(-5) - 6(-3) = -17$$
$$c = (-3)(-2) - (-5)(-4) = -14$$

Using the point $A(4,2,-3)$,

$$-10x - 17y - 14z = d$$
$$d = (-10)(4) + (-17)(2) + (-14)(-3) = -32$$

Therefore, the equation of the plane is

$$-10x - 17y - 14z = -32$$

$$10x + 17y + 14z = 32$$

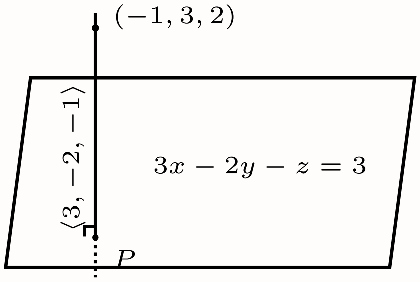
Example 9

Find the equation of the line through the point $(-1,3,2)$ and perpendicular to the plane $3x-2y-z=3$. Find the point of intersection of the line and plane.

Solution

Directed values of the line perpendicular to the plane $3x-2y-z=3$ are $\langle 3,-2,-1\rangle$. Then the equation of the line is

$$\frac{x-(-1)}{3}=\frac{y-3}{-2}=\frac{z-2}{-1}$$
$$\frac{x+1}{3}=\frac{y-3}{-2}=\frac{z-2}{-1}$$



So, coordinates of the points on this line are

$(x,y,z)=(-1+3k,\ 3-2k,\ 2-k),$ for some real number k.

If one of these points P lies on the plane, then

$$\begin{aligned}
3(-1+3k)-2(3-2k)-(2-k)&=3,\\
-3+9k-6+4k-2+k&=3,\\
14k-11&=3,\\
14k&=14,\\
k&=1.
\end{aligned}$$

Hence, the point of intersection of the line and the given plane is

$$P=(x,y,z)=(-1+3(1),\ 3-2(1),\ 2-1)=(2,1,1).$$

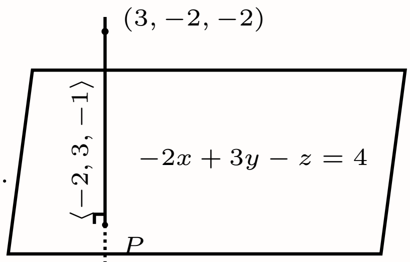
Exercise 3.4

2. Find the equation of the line that passes through the point $(3,-2,-2)$ and perpendicular to the plane $-2x+3y-z=4$. Find the point of intersection of the line and the given plane.

Solution

Directed values of the line perpendicular to the plane $-2x+3y-z=4$ are $\langle -2,3,-1\rangle$. Then the equation of the line is

$$\frac{x-3}{-2}=\frac{y+2}{3}=\frac{z+2}{-1}$$



So, coordinates of the points on this line are

$(x,y,z)=(3-2k,\ -2+3k,\ -2-k),$ for some real number k.

If one of these points P lies on the plane, then

$$\begin{aligned}
-2(3-2k)+3(-2+3k)-(-2-k)&=4,\\
-6+4k-6+9k+2+k&=4,\\
14k-10&=4,\\
14k&=14,\\
k&=1.
\end{aligned}$$

Hence, the point of intersection of the line and the given plane is

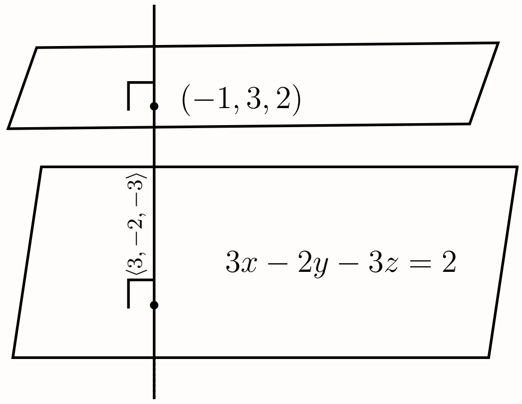
$$P=(x,y,z)=(3-2(1),\ -2+3(1),\ -2-1)=(1,1,-3).$$

Example 10

Find the equation of the plane containing the point $(-1, 3, 2)$ and parallel to the plane $3x - 2y - 3z = 2$.

Solution (1)

Since the required plane is parallel to the plane $3x - 2y - 3z = 2$, both planes have the same normal vector $\langle 3, -2, -3\rangle$.



Therefore, the equation of the required plane is of the form

$$3x - 2y - 3z = d$$

Using the point $(-1, 3, 2)$,

$$d = 3(-1) - 2(3) - 3(2) = -15$$

Therefore, the equation of the plane is

$$3x - 2y - 3z = -15$$

Solution (2)

Directed values of the line perpendicular to the given plane $3x - 2y - 3z = 2$ are $\langle 3, -2, -3\rangle$. This line is also perpendicular to the required plane. So the equation of the required plane is

$$3x - 2y - 3z = d$$

Since the point $(-1, 3, 2)$ is on the plane,

$$d = 3(-1) - 2(3) - 3(2) = -15$$

The equation of the required plane is

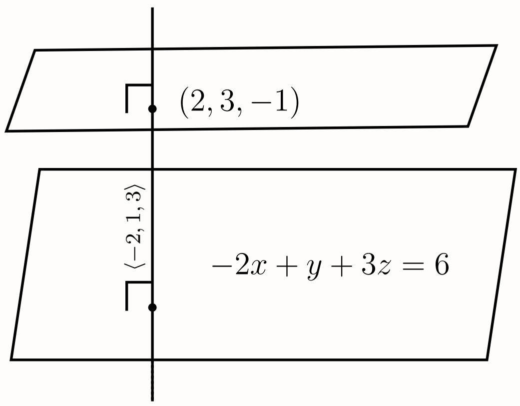
$$3x - 2y - 3z = -15$$

Exercise 3.4

3. Find the equation of the plane containing the point $(2, 3, -1)$ and parallel to the plane $-2x + y + 3z = 6$.

Solution

Directed values of the line perpendicular to the given plane $-2x+y+3z = 6$ are $\langle -2, 1, 3\rangle$. This line is also perpendicular to the required plane. So the equation of the required plane is



$$-2x + y + 3z = d$$

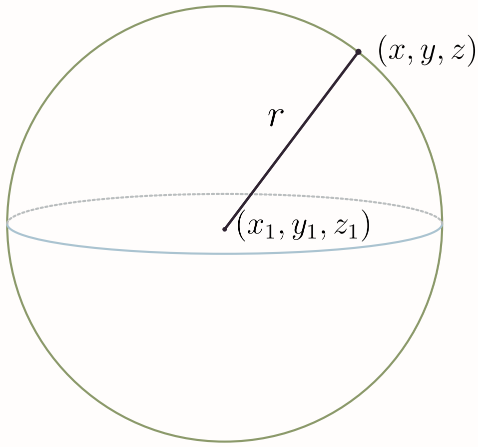
Since the point $(2, 3, -1)$ is on the plane,

$$d = -2(2) + 3 + 3(-1) = -4$$

The equation of the required plane is

$$-2x + y + 3z = -4$$

3.5 Spheres



The distance between center (x_1, y_1, z_1) of a sphere and any point (x, y, z) on the sphere is radius r.

$$\sqrt{(x-x_1)^2+(y-y_1)^2+(z-z_1)^2}=r$$

Therefore, the equation of the sphere with center (x_1, y_1, z_1) and radius r is

$$\boxed{(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=r^2}$$

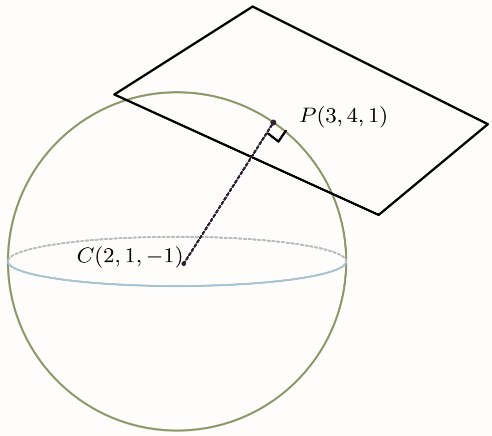
Example 11

Find the equation of the plane tangent to the sphere $(x-2)^2+(y-1)^2+(z+1)^2=14$ at the point $(3, 4, 1)$.

Solution

The center of the sphere is $C(2, 1, -1)$.
Directed values of the line joining the center $(2, 1, -1)$ of the sphere and the given point $(3, 4, 1)$ are $\langle 1, 3, 2\rangle$.
This line is perpendicular to the tangent plane.
Therefore, the equation of the tangent plane is

$$x+3y+2z=d$$



Since the point $(3, 4, 1)$ is on the tangent plane,

$$d=3(3)+3(4)+2(1)=17$$

Therefore, the equation of the tangent plane is

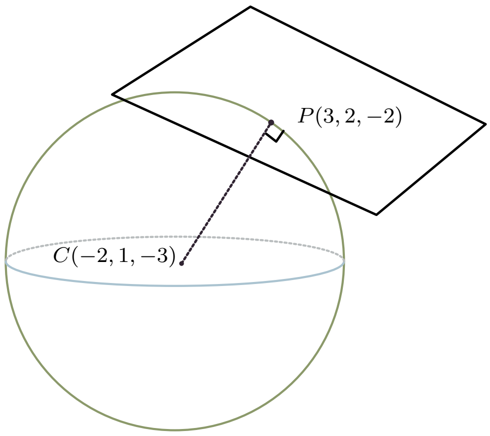
$$x+3y+2z=17$$

Exercise 3.5

4. Find the equation of the plane tangent to the sphere $(x+2)^2+(y-1)^2+(z+3)^2=27$ at the point $(3,2,-2)$.

Solution

The center of the sphere is $C(-2,1,-3)$.
Directed values of the line joining the center $(-2,1,-3)$ of the sphere and the given point $(3,2,-2)$ are $\langle 5,1,1\rangle$.
This line is perpendicular to the tangent plane.
Therefore, the equation of the tangent plane is



$$5x+y+z=d$$

Since the point $(3,2,-2)$ is on the tangent plane,

$$d=5(3)+2+(-2)=15$$

Therefore, the equation of the tangent plane is

$$5x+y+z=15$$

Example 12

Find the equation of the sphere with center $(0,1,0)$ and touching the plane $x-2y+2z+5=0$.

Solution

Directed values of the line (radius) joining the center $(0,1,0)$ of the sphere and perpendicular to the plane $x-2y+2z+5=0$ are $\langle 1,-2,2\rangle$.
Coordinates of the point on this line (radius) are

$$(x,y,z)=(k,1-2k,2k) \quad \text{for some real number } k.$$

If one of these points (x,y,z) is on the plane, then

$$k-2(1-2k)+2(2k)+5=0 \Rightarrow 9k+3=0 \Rightarrow k=-\frac{1}{3}$$

Therefore, the touching point is

$$\left(-\frac{1}{3},1-2\left(-\frac{1}{3}\right),2\left(-\frac{1}{3}\right)\right)=\left(-\frac{1}{3},\frac{5}{3},-\frac{2}{3}\right)$$

Hence the radius is

$$r=\sqrt{\left(-\frac{1}{3}\right)^2+\left(\frac{5}{3}-1\right)^2+\left(-\frac{2}{3}\right)^2}=1$$

Therefore, the equation of the sphere is

$$x^2+(y-1)^2+z^2=1$$

5. Find the equation of the sphere with center $(6,-7,-3)$ and touching the plane $4x-2y-z=17$.

Solution

Directed values of the line (radius) joining the center $(6,-7,-3)$ of the sphere and perpendicular to the touching plane $4x-2y-z=17$ are $\langle 4,-2,-1\rangle$.
Coordinates of the point on this line (radius) are

$$(x,y,z)=(6+4k,-7-2k,-3-k) \quad \text{for some real number } k.$$

If one of these points (x,y,z) is on the plane, then

$$4(6+4k)-2(-7-2k)-(-3-k)=17 \Rightarrow 41+21k=17 \Rightarrow k=-\frac{8}{7}$$

Therefore, the touching point is

$$\left(6+4\left(-\frac{8}{7}\right),-7-2\left(-\frac{8}{7}\right),-3-\left(-\frac{8}{7}\right)\right)=\left(\frac{10}{7},-\frac{33}{7},-\frac{13}{7}\right)$$

Hence the radius is

$$r=\sqrt{\left(\frac{10}{7}-6\right)^2+\left(-\frac{33}{7}+7\right)^2+\left(-\frac{13}{7}+3\right)^2}=\sqrt{\frac{64}{21}}$$

Therefore, the equation of the sphere is

$$(x-6)^2+(y+7)^2+(z+3)^2=\frac{64}{21}$$

3. Find the equation of the sphere on the join of $(1,-1,1)$ and $(-3,4,5)$ as diameter.

Solution

Let $A(1,-1,1)$ and $B(-3,4,5)$.
The center of the sphere is the midpoint of the ends of the diameter. Therefore,

$$C=\left(\frac{1+(-3)}{2},\frac{-1+4}{2},\frac{1+5}{2}\right)=\left(-1,\frac{3}{2},3\right)$$

Directed values of the diameter joining $(1,-1,1)$ and $(-3,4,5)$ are $\langle -4,5,4\rangle$. Hence,

$$AB=\sqrt{(-4)^2+5^2+4^2}=\sqrt{57}$$

Therefore, the radius is

$$r=\frac{\sqrt{57}}{2}$$

So, the equation of the sphere is

$$(x+1)^2+\left(y-\frac{3}{2}\right)^2+(z-3)^2=\frac{57}{4}$$

1. Find the equation of the sphere with center C and radius r,
   (a) $C(1,-2,4),\ r=3$
   (b) $C(2,6,-3),\ r=2$
   (c) $C(2,3,5),\ r=5$

   Solution

   (a) The equation of the sphere with center $C(1,-2,4)$ and radius 3 is

$$(x-1)^2+(y+2)^2+(z-4)^2=9$$

   (b) The equation of the sphere with center $C(2,6,-3)$ and radius 2 is

$$(x-2)^2+(y-6)^2+(z+3)^2=4$$

   (c) The equation of the sphere with center $C(2,3,5)$ and radius 5 is

$$(x-2)^2+(y-3)^2+(z-5)^2=25$$

2. Check whether the given point P lies inside, outside or on a sphere.
   (a) $C(0,0,0),\ r=3$ and $P(1,1,1)$
   (b) $C(0,0,0),\ r=3$ and $P(2,1,2)$
   (c) $C(0,0,0),\ r=3$ and $P(10,10,10)$

   Solution

   (a) $CP=\sqrt{1^2+1^2+1^2}=\sqrt{3}$. Since $CP<r$, the point $P(1,1,1)$ lies inside the sphere.
   (b) $CP=\sqrt{2^2+1^2+2^2}=\sqrt{9}=3$. Since $CP=r$, the point $P(2,1,2)$ lies on the sphere.
   (c) $CP=\sqrt{10^2+10^2+10^2}=\sqrt{300}=10\sqrt{3}$. Since $CP>r$, the point $P(10,10,10)$ lies outside the sphere.

6. What is the equation of the sphere which passes through the points $(3,0,2)$, $(-1,1,1)$ and $(2,-5,4)$ and whose center lies on the plane $2x+3y+4z=6$?

   Solution

   Let the center of the sphere be $C(x_1,y_1,z_1)$ and the radius be r. Then the equation of the sphere is

$$(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=r^2$$

   Since the sphere passes through the points $(3,0,2)$, $(-1,1,1)$ and $(2,-5,4)$, we have

$$\begin{aligned}(3-x_1)^2+(0-y_1)^2+(2-z_1)^2&=r^2\\ x_1^2-6x_1+9+y_1^2+z_1^2-4z_1+4&=r^2 \qquad (1)\end{aligned}$$

$$\begin{aligned}(-1-x_1)^2+(1-y_1)^2+(1-z_1)^2&=r^2\\ x_1^2+2x_1+1+y_1^2-2y_1+1+z_1^2-2z_1+1&=r^2 \qquad (2)\end{aligned}$$

$$\begin{aligned}(2-x_1)^2+(-5-y_1)^2+(4-z_1)^2&=r^2\\ x_1^2-4x_1+4+y_1^2+10y_1+25+z_1^2-8z_1+16&=r^2 \qquad (3)\end{aligned}$$

   Also, since the center lies on the plane $2x+3y+4z=6$,

$$2x_1+3y_1+4z_1=6 \qquad (4)$$

   Subtracting (2) from (1), we get

$$-6x_1 + 9 - 4z_1 + 4 = 2x_1 + 1 - 2y_1 + 1 - 2z_1 + 1$$
$$-8x_1 + 2y_1 - 2z_1 + 10 = 0$$
$$-4x_1 + y_1 - z_1 = -5 \tag{5}$$

Multiplying (4) by 2, we get

$$4x_1 + 6y_1 + 8z_1 = 12$$

Adding this to (5), x_1 is eliminated. Therefore,

$$7y_1 + 7z_1 = 7$$
$$y_1 + z_1 = 1 \tag{6}$$

Subtracting (2) from (3), we get

$$-4x_1 + 4 + 10y_1 + 25 - 8z_1 + 16 = 2x_1 + 1 - 2y_1 + 1 - 2z_1 + 1$$
$$-6x_1 + 12y_1 - 6z_1 + 42 = 0$$
$$x_1 - 2y_1 + z_1 = 7 \tag{7}$$

Multiplying (7) by 2 and subtracting it from (4), x_1 is eliminated. Therefore,

$$2x_1 + 3y_1 + 4z_1 - (2x_1 - 4y_1 + 2z_1) = 6 - 14$$
$$7y_1 + 2z_1 = -8 \tag{8}$$

Multiplying (6) by 2 and subtracting it from (8), we get

$$7y_1 + 2z_1 - (2y_1 + 2z_1) = -8 - 2$$
$$5y_1 = -10, \quad y_1 = -2$$
$$z_1 = 3 \quad \text{from (6)}$$
$$x_1 = 0 \quad \text{from (7)}$$

Since, the center of the sphere is $C(0, -2, 3)$. Substituting in (1),

$$(3-0)^2 + (0+2)^2 + (2-3)^2 = r^2$$
$$r^2 = 14$$
$$r = \sqrt{14}$$

Therefore, the equation of the sphere is

$$x^2 + (y+2)^2 + (z-3)^2 = 14$$

Example 13

Find the equation of a sphere that passes through the points $(9, 0, 0)$, $(3, 13, 5)$ and $(11, 0, 10)$, given that its center lies on the yz-plane.

Solution

Since the center lies on the yz-plane, let the center of the sphere be $C(0, y_1, z_1)$ and the radius be r. Then the equation of the sphere is

$$x^2 + (y - y_1)^2 + (z - z_1)^2 = r^2$$

Since the sphere passes through the points $(9, 0, 0)$, $(3, 13, 5)$ and $(11, 0, 10)$, we have

$$9^2 + y_1^2 + z_1^2 = r^2$$

$$81 + y_1^2 + z_1^2 = r^2 \qquad (1)$$

$$3^2 + (13 - y_1)^2 + (5 - z_1)^2 = r^2$$

$$203 - 26y_1 - 10z_1 + y_1^2 + z_1^2 = r^2 \qquad (2)$$

$$11^2 + y_1^2 + (10 - z_1)^2 = r^2$$

$$221 + y_1^2 - 20z_1 + z_1^2 = r^2 \qquad (3)$$

Subtracting (1) from (2), we get

$$122 - 26y_1 - 10z_1 = 0$$

$$13y_1 + 5z_1 = 61 \qquad (4)$$

Subtracting (1) from (3), we get

$$140 - 20z_1 = 0$$

$$z_1 = 7$$

Substituting this in (4), we get

$$13y_1 + 5(7) = 61$$

$$13y_1 = 26$$

$$y_1 = 2$$

Therefore, the center of the sphere is $C(0, 2, 7)$. Substituting in (1),

$$81 + 2^2 + 7^2 = r^2$$

$$r^2 = 134$$

$$r = \sqrt{134}$$

Therefore, the equation of the sphere is

$$x^2 + (y - 2)^2 + (z - 7)^2 = 134$$